Binomial Expansions as Sets

The Combinatorial Interpretation of Polynomials

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M1 Extended notes for F.4 Binomial Theorem

Deconstructing Algebraic Forms

The Intuition Behind ax^b

Let us re-examine basic polynomial elements through a set-theoretic lens:

- **Addition ($+$):** Counts and collects distinct varieties.
- **Multiplication ($\times$):** Assembles composite multi-sets.

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The Linear Form as a Database

The expression $ax + b$ is a **compressed set representation**:

- It holds a selectable copies of object x.
- It holds b baseline identity markers (1).

The Mechanics of Expansion

Consider the n-th power of a system: $(ax + by)^n$

- Expanding this means drawing exactly one variable component from each of the n available independent brackets.
- To construct a combined outcome state matching $x^k y^{n-k}$:

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Combinatorial Selection

We must choose the trait x from exactly k parent brackets, leaving the remaining $n - k$ brackets to provide y. The total paths to achieve this matches the combination formula:

$$\text{Total Configuration Paths} = C(n, k) = \binom{n}{k}$$

Polynomial Expansions and Event Valuations

What happens if we expand a multi-option system $(a + bx + cx^2 + \dots)^n$?

- The exponents (x^0, x^1, x^2) serve as tags for physical scores or event values.
- **Valuation Rule:** $\mathrm{val}(x^i y^j) = i + j$.

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- **Valuation Rule:** $\mathrm{val}(x^i y^j) = i + j$.
- Because algebraic multiplication triggers exponent addition, expanding polynomials automatically sums up the cumulative scores across multi-stage trials.

Application: Modeling Dice Rolls

A fair, symmetric 6-sided die can be written as a choice polynomial:

$$S(x) = x + x^2 + x^3 + x^4 + x^5 + x^6 = \sum_{i=1}^{6} x^i$$

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Joint Distribution Laws

When rolling n independent dice:

- The entire score tracking grid is contained within $S(x)^n$.
- The number of ways to achieve a total score sum of r is:

$$\text{Ways}(r) = \mathrm{coef}\left(x^r \text{ in } S(x)^n\right)$$

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Probability Formula

$$P(\text{Sum} = r) = \frac{\mathrm{coef}\left(x^r \text{ in } S(x)^n\right)}{6^n}$$

Summary: Structural Alignments

Algebraic Element	**Combinatorial Meaning**
Coefficient (a)	Frequency / Selection Paths
Exponent (b)	Structural Valuation / Score
Multiplication ($\times$)	Composition of Multi-Stage Events
Polynomial Product ($S(x)^n$)	Complete Joint Sample Space

Table: Mapping Algebra to Discrete Probability Spaces